

Solution Problem 3.2

$$\text{Assume } \begin{cases} Z \sim N(0, 1) \\ Y \sim \chi^2_{(n-1)} \end{cases}$$

$$\text{Then } X \stackrel{\text{def}}{=} \frac{Z}{\sqrt{Y/(n-1)}} \stackrel{\text{def}}{\sim} t(n-1)$$

$$\text{Also } U \stackrel{\text{def}}{=} Z^2 \stackrel{\text{def}}{\sim} \chi^2(1)$$

$$\text{and } V \stackrel{\text{def}}{=} \frac{U/1}{Y/(n-1)} \stackrel{\text{def}}{\sim} F(1, n-1)$$

Since $X^2 = V$ it follows that

$$\boxed{X \sim t(n-1) \Rightarrow X^2 \sim F(1, n-1)} \quad \text{QED.}$$

Solution Problem 3.3 (see also m-file)

$$X \sim N_p(\mu, \Sigma) \quad \mu, \Sigma \text{ unknown}$$

Observations $X_1, \dots, X_n$ iid $\sim N_p(\mu, \Sigma)$

$$\text{Likelihood Ratio test } \begin{aligned} H_0: \Sigma &= \Sigma_0 \\ H_1: \Sigma &\neq \Sigma_0 \end{aligned}$$

Given in text

$$\begin{aligned} -2 \log \Lambda &= -2 \log \frac{\max_{\mu} L(\mu, \Sigma_0)}{\max_{\mu, \Sigma} L(\mu, \Sigma)} \\ &= n \text{trace}(\Sigma_0^{-1} S) - n \log |\Sigma_0^{-1} S| - np \end{aligned}$$

$$\text{df}\{L(\mu, \Sigma_0)\} = p \quad (\text{for } \mu)$$

$$\begin{aligned} \text{df}\{L(\mu, \Sigma)\} &= p + p + [1 + 2 + \dots + (p-1)] \\ &= p + p + \frac{1}{2}(p-1)p \\ &= p + \frac{1}{2}p(p+1) \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

$$\boxed{\text{Test: Reject } H_0 \text{ if } -2 \log \Lambda \geq \chi^2_{\frac{1}{2}p(p+1)}(\alpha)}$$

Solution Problem 3.4a (see also m-file)

$X \sim \text{Poisson}(\lambda)$ $f_X(x) = \frac{\lambda^x}{x!} e^{-\lambda}$, $x=0,1,\dots$ λ unknown

Observations $X_1, \dots, X_n$ iid

Likelihood function $L(\text{obs}|\lambda) = \prod f_X(x_i) = \frac{\lambda^{\sum x_i}}{\prod x_i!} e^{-n\lambda}$

ML estimate: $\log L = \sum x_i \log \lambda - n\lambda - \log \prod x_i!$

$$\frac{\partial \log L}{\partial \lambda} = \frac{\sum x_i}{\lambda} - n = 0$$
$$\Downarrow$$
$$\hat{\lambda}_{ML} = \bar{x}$$

$$H_0: \lambda = \lambda_0$$
$$H_1: \lambda \neq \lambda_0$$

$$\text{Likelihood ratio } \Lambda = \frac{\max_{\lambda} L(\text{obs}|\lambda)}{\max_{\lambda} L(\text{obs}|\lambda)} \leftarrow df=0$$
$$= \frac{\lambda_0^{\sum x_i} e^{-n\lambda_0}}{\prod x_i!} \cdot \frac{\prod x_i!}{\bar{x}^{\sum x_i} e^{-n\bar{x}}}$$
$$= \left(\frac{\lambda_0}{\bar{x}}\right)^{n\bar{x}} e^{n(\bar{x} - \lambda_0)}$$

Large sample LR-test

$$-2 \log \Lambda = -2n\bar{x} \log \frac{\lambda_0}{\bar{x}} - 2n(\bar{x} - \lambda_0)$$

Since $df_{L(\lambda_0)} = 0$, $df_{L(\lambda)} = 1$

$$-2 \log \Lambda \stackrel{H_0}{\sim} \chi_1^2 \text{ (approximately)}$$

Test: Reject H_0 if

$$2n \left(\lambda_0 - \bar{x} - \bar{x} \log \frac{\lambda_0}{\bar{x}} \right) \geq \chi_1^2(\alpha)$$

Solution Problem 3.4b (see also m-file)

$$T \sim \text{Exp}(\lambda) \quad f_T(t|\lambda) = \lambda e^{-\lambda t} \quad (\lambda \text{ unknown})$$

Observations $t_1, \dots, t_n$ iid $\sim \text{Exp}(\lambda)$

$$\begin{aligned} \text{Likelihood function } L(\text{obs}|\lambda) &= \prod_{i=1}^n f_T(t_i|\lambda) \\ &= \lambda^n e^{-\lambda \sum t_i} \end{aligned}$$

$$\begin{aligned} \text{ML estimate: } \frac{\partial L}{\partial \lambda} &= n\lambda^{n-1} e^{-\lambda \sum t_i} - \lambda^n \sum t_i e^{-\lambda \sum t_i} = 0 \\ \Downarrow \\ \hat{\lambda}_{ML} &= \frac{n}{\sum t_i} = 1/\bar{t} \end{aligned}$$

$$H_0: \lambda = \lambda_0$$

$$H_1: \lambda \neq \lambda_0$$

$$\begin{aligned} \text{Likelihood ratio } \Lambda &= \frac{\max_{\lambda} L(\text{obs}|\lambda_0)}{\max_{\lambda} L(\text{obs}|\lambda)} \quad \leftarrow df=0 \\ &= \frac{\lambda_0^n e^{-\lambda_0 \sum t_i}}{\hat{\lambda}_{ML}^n e^{-\hat{\lambda}_{ML} \sum t_i}} \quad \leftarrow df=1 \\ &= (\lambda_0 \bar{t})^n e^{n(1-\lambda_0 \bar{t})} \end{aligned}$$

Large sample LR-test:

$$-2 \log \Lambda = -2n \log \lambda_0 \bar{t} - 2n(1 - \lambda_0 \bar{t})$$

Since $df_{L(\lambda_0)} = 0$, $df_{L(\lambda)} = 1$

$$-2 \log \Lambda \sim \chi^2_1 \quad (\text{approximately})$$

Test: Reject H_0 if

$$2n(\lambda_0 \bar{t} - \log \lambda_0 \bar{t} - 1) \geq \chi^2_1(\alpha)$$